APG-AMS: Entropy-Guided Screening for Adaptive Metric Search in Distance-Based Learning

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Abstract

Distance-based learning algorithms depend critically on the choice of metric. Exhaustive search over candidate Minkowski norms may improve local neighborhood behavior, but it also increases computational cost through repeated distance evaluations. This paper introduces Arithmetic Power Geometry Adaptive Metric Selection (APG–AMS), an entropy-guided screening framework for reducing unnecessary adaptive metric search.

The method is motivated by a local APG exponent-deformation result: near the Euclidean reference exponent $p = 2$, the first-order sensitivity of the normalized power-sum distortion is governed by the Shannon entropy of the squared-coordinate weight distribution. APG–AMS uses normalized APG entropy to identify potentially metric-sensitive query samples and applies full local p-search only when the entropy criterion predicts elevated metric sensitivity.

Experiments were conducted on eight benchmark and controlled datasets using 10-fold stratified cross-validation. At threshold $\tau = 0.92$, APG–AMS produced predictive accuracy statistically indistinguishable from both Euclidean k-nearest neighbors and full adaptive p-search. The primary evaluation used 10-fold stratified cross-validation, while the earlier 70/30 split experiment is reported only as an auxiliary threshold-ablation study.

Across 80 paired folds, APG–AMS versus Euclidean k-NN yielded paired t-test $p = 0.417$ and Wilcoxon $p = 0.289$, while APG–AMS versus full p-search yielded paired t-test $p = 0.864$ and Wilcoxon $p = 0.641$. The results suggest that APG entropy can serve as a practical screening indicator for metric stability in the evaluated datasets. The proposed framework is not intended as a replacement for supervised metric-learning methods, but rather as a computational screening layer for adaptive metric search.

Keywords: Arithmetic Power Geometry; adaptive metric selection; k-nearest neighbors; Shannon entropy; information geometry; metric learning; Minkowski distance.

1 Introduction

Distance is a central computational primitive in machine learning. Nearest-neighbor classification, clustering, manifold learning, retrieval, and similarity search all depend on the geometry induced by a chosen metric (Fix and Hodges, 1951; Cover and Hart, 1967; Duda et al., 2001; Bishop, 2006; Goodfellow et al., 2016; Hastie et al., 2009; Murphy, 2012; Zhang, 2016). The classical k-nearest-neighbor (k-NN) classifier is conceptually simple: a query point is classified from the labels of the closest training samples. Its performance, however, can vary significantly with the distance function (Cover and Hart, 1967; Devroye et al., 1996; Cunningham and Delany, 2021). For this reason, metric learning and metric selection remain important research topics (Xing et al., 2003; Goldberger et al., 2005; Weinberger and Saul, 2009; Kulis, 2013; Bellet et al., 2013).

A common approach is to search over candidate Minkowski norms, such as L_1, L_2, L_3, or fractional L_p distances. Full local metric search can be useful, but it increases the number of distance computations by a factor equal to the number of tested metrics. Learned metric methods, including Neighborhood Components Analysis (NCA) and Large Margin Nearest Neighbor (LMNN), address metric selection through optimization, but they introduce their own training cost and model assumptions (Goldberger et al., 2005; Weinberger and Saul, 2009; Kulis, 2013). In many practical settings, a simpler question is useful: *when is metric adaptation unnecessary?*

Arithmetic Power Geometry (APG) offers a way to frame this question. APG treats the norm exponent as a continuous deformation parameter rather than as a fixed discrete choice (Akhtar, 2026a,b). In the local APG setting, the Euclidean exponent $p = 2$ is taken as the reference state, and small perturbations away from this state are studied analytically. The key observation is that the first-order sensitivity of a vector under exponent deformation is governed by the Shannon entropy of the squared-coordinate weight distribution (Shannon, 1948; Cover and Thomas, 2006; Amari, 2016; Nielsen, 2020). This suggests that entropy can be used as a screening variable for metric stability.

This paper develops APG-AMS, an entropy-guided adaptive metric-selection framework. APG-AMS classifies low-entropy query samples as metric-stable and uses Euclidean distance for them. It applies full local p-search only to high-entropy samples, where APG predicts greater sensitivity to exponent deformation. The purpose is not to outperform all classifiers. Instead, the goal is to reduce unnecessary metric-search operations while preserving predictive performance.

The paper makes four contributions. First, it states and proves a d-dimensional APG local sensitivity theorem for normalized power-sum distortion. Second, it formulates APG-AMS as a reproducible metric-search screening algorithm. Third, it evaluates APG-AMS across eight datasets using 10-fold stratified cross-validation, threshold ablation, and runtime scaling. Fourth, it reports paired statistical tests showing no significant accuracy loss relative to Euclidean k-NN or full adaptive p-search in the evaluated setting.

2 Background and Related Work

2.1 Nearest-neighbor classification and metric dependence

The k-NN rule is one of the earliest and most widely used nonparametric classification methods (Fix and Hodges, 1951; Cover and Hart, 1967). Its asymptotic behavior is well studied (Devroye et al., 1996), and it remains widely used in applied machine learning because of its interpretability and lack of explicit training phase (Bishop, 2006; Hastie et al., 2009; Cunningham and Delany, 2021). The method is nevertheless highly metric-dependent. A poor metric may distort neighborhoods and reduce classification performance.

2.2 Metric learning

Metric learning attempts to learn a distance function from data. NCA optimizes a stochastic leave-one-out neighbor assignment objective (Goldberger et al., 2005). LMNN learns a Mahalanobis metric intended to keep target neighbors close while pushing differently labeled samples away (Weinberger and Saul, 2009). Broader surveys emphasize that metric learning can improve performance but requires additional optimization, parameter selection, and validation (Kulis, 2013; Bellet et al., 2013). APG-AMS is different in purpose. It does not learn a full metric. It screens query samples to decide whether local metric search should be performed.

2.3 Entropy and information geometry

Entropy measures uncertainty or dispersion in a probability distribution (Shannon, 1948; Renyi, 1961; Cover and Thomas, 2006). Information geometry studies statistical models using differential-geometric tools (Amari and Nagaoka, 2000; Amari, 2016; Ay et al., 2017; Nielsen, 2020). APG-AMS uses an entropy derived from the squared coordinates of a feature vector. This entropy is not a class entropy; it is a coordinate-dispersion measure. Its use is justified by the local exponent-deformation calculation in Section 3.

3 Theory

Let $x = (x_1, \ldots, x_d) \in \mathbb{R}^d_{>0}$ be a positive feature vector and define

$$S_2(x) = \sum_{i=1}^{d} x_i^2. \tag{1}$$

For $p > 0$, define the normalized power-sum ratio

$$R_p(x) = \frac{\sum_{i=1}^{d} x_i^p}{\left(\sum_{i=1}^{d} x_i^2\right)^{p/2}}. \tag{2}$$

At $p = 2$, $R_2(x) = 1$. Let

$$K_P = p - 2 \tag{3}$$

be the local exponent-deformation parameter. Define squared-coordinate weights

$$w_i = \frac{x_i^2}{\sum_{j=1}^{d} x_j^2}, \qquad i = 1, \ldots, d, \tag{4}$$

so that $\sum_i w_i = 1$. The unnormalized APG entropy is

$$H(W) = -\sum_{i=1}^{d} w_i \log w_i. \tag{5}$$

For cross-dimensional comparison, the algorithm uses the normalized entropy

$$\widehat{H}(W) = \frac{H(W)}{\log d}, \tag{6}$$

when $d > 1$.

Theorem 1 (APG local entropy sensitivity)**.** *For fixed $x \in \mathbb{R}^d_{>0}$ and $p = 2 + K_P$ with $K_P \to 0$,*

$$R_p(x) = 1 - \frac{H(W)}{2} K_P + O(K_P^2). \tag{7}$$

Consequently, the normalized local distortion

$$\delta_K(x, p) = |1 - R_p(x)| \tag{8}$$

satisfies

$$\delta_K(x, p) = \frac{H(W)}{2} |K_P| + O(K_P^2). \tag{9}$$

Proof. Let $p = 2 + K_P$. Using $x_i^{2+K_P} = x_i^2 \exp(K_P \log x_i)$,

$$\sum_{i=1}^{d} x_i^{2+K_P} = S_2(x) + K_P \sum_{i=1}^{d} x_i^2 \log x_i + O(K_P^2). \tag{10}$$

Also,

$$S_2(x)^{1+K_P/2} = S_2(x) \left(1 + \frac{K_P}{2} \log S_2(x) + O(K_P^2)\right). \tag{11}$$

Therefore,

$$\begin{aligned} R_{2+K_P}(x) &= \left(1 + K_P \frac{\sum_i x_i^2 \log x_i}{S_2(x)}\right)\left(1 - \frac{K_P}{2} \log S_2(x)\right) + O(K_P^2) &(12)\\ &= 1 + K_P \left[\frac{\sum_i x_i^2 \log x_i}{S_2(x)} - \frac{1}{2} \log S_2(x)\right] + O(K_P^2). &(13) \end{aligned}$$

Since

$$\frac{x_i^2}{S_2(x)} = w_i, \tag{14}$$

we have

$$\begin{aligned} \frac{\sum_i x_i^2 \log x_i}{S_2(x)} - \frac{1}{2} \log S_2(x) &= \frac{1}{2} \sum_i w_i \log\left(\frac{x_i^2}{S_2(x)}\right) &(15)\\ &= \frac{1}{2} \sum_i w_i \log w_i &(16)\\ &= -\frac{H(W)}{2}. &(17) \end{aligned}$$

Thus

$$R_{2+K_P}(x) = 1 - \frac{H(W)}{2} K_P + O(K_P^2). \tag{18}$$

Taking absolute deviation from one gives the stated local distortion relation. □

Remark 1. *The theorem is local near $p = 2$. It does not claim that APG entropy globally replaces exact Minkowski distances or learned metric methods. It supports entropy as a first-order screening indicator for local metric sensitivity.*

Important Scope Clarification. Theorem 1 is used only as a local sensitivity indicator near the Euclidean reference exponent $p = 2$. APG–AMS does not assume that the first-order expansion in Equation (18) remains quantitatively accurate for large exponent deviations such as $p = 1$ or $p = 4$. The adaptive search stage is empirical and uses Theorem 1 only as a motivation for entropy-guided screening.

4 Methodology

4.1 APG-AMS algorithm

For each test query x, APG-AMS computes $\widehat{H}(W_x)$. Given a threshold $\tau \in [0, 1]$, it applies the following rule:

$$\text{metric}(x) = \begin{cases} L_2, & \widehat{H}(W_x) < \tau, \\ \arg\max_{p \in \{1,1.5,2,3,4\}} \text{local purity}_k(x; p), & \widehat{H}(W_x) \geq \tau. \end{cases} \tag{19}$$

The local purity score is the proportion of the majority class among the k nearest neighbors for a candidate p. Local purity is used as a practical neighborhood-quality criterion and is not claimed to be an optimal metric-selection objective. The APG–AMS framework requires only a local ranking criterion and does not depend on the optimality of local purity. In all experiments, $k = 7$ and the candidate metric set is

$$\mathcal{P} = \{1, 1.5, 2, 3, 4\}. \tag{20}$$

A full adaptive search evaluates all five metrics for every query. APG-AMS evaluates only L_2 for entropy-stable queries and evaluates all five metrics only for entropy-sensitive queries.

4.2 Datasets

Eight datasets were evaluated: Iris, Wine, Breast Cancer Wisconsin Diagnostic, Digits, Synthetic-20D, Synthetic-50D, Two Moons, and Concentric Circles. The first four are standard benchmark datasets available through widely used machine-learning libraries and repositories (Fisher, 1936; Dua and Graff, 2019; Pedregosa et al., 2011). The remaining four are controlled synthetic datasets generated for stress testing nonlinear and high-dimensional behavior (Pedregosa et al., 2011).

4.3 Preprocessing and evaluation protocol

All features were standardized within each training fold using the training data only. Entropy was computed after min-max mapping the standardized query representation into a positive interval, because APG weights require nonnegative squared-coordinate mass. The core evaluation used 10-fold stratified cross-validation with threshold $\tau = 0.92$. Accuracy and weighted F1-score were recorded. Search reduction was computed as

$$\text{Reduction} = 100\left(1 - \frac{\text{APG searches}}{\text{full adaptive searches}}\right). \tag{21}$$

Paired t-tests and Wilcoxon signed-rank tests were used to compare APG-AMS accuracy against Euclidean k-NN and full adaptive p-search across the 80 dataset-fold pairs (Student, 1908; Wilcoxon, 1945; Demšar, 2006). Numerical implementation used NumPy, SciPy, pandas, and scikit-learn (Harris et al., 2020; Virtanen et al., 2020; McKinney, 2010; Pedregosa et al., 2011).

5 Results

5.1 Ten-fold cross-validation

Table 1 reports mean accuracy and search reduction across 10 folds.

Table 1: Ten-fold cross-validation summary at $\tau = 0.92$.

Dataset	L_2 Acc.	Full Acc.	APG Acc.	Search Red.
Breast Cancer	96.488	96.310	96.313	76.348
Concentric Circles	93.400	93.100	93.400	52.320
Digits	97.439	97.774	97.439	80.000
Iris	96.000	94.000	94.667	49.600
Synthetic-20D	93.300	93.600	93.500	10.880
Synthetic-50D	72.400	72.900	72.900	0.240
Two Moons	94.500	93.900	94.100	60.160
Wine	96.667	97.222	96.667	69.725

The results show that APG–AMS maintained accuracy close to both Euclidean k-NN and full p-search.

Search reduction varied substantially across datasets. It was strongest for Digits, Breast Cancer, Wine, Two Moons, and Concentric Circles.

The Synthetic-50D benchmark provides an informative failure mode. For this dataset the search reduction was approximately 0.24%, indicating that APG–AMS effectively converged to full adaptive search.

This behavior is theoretically consistent with the APG entropy interpretation. In high-dimensional balanced feature spaces, normalized entropy approaches its maximum value and most query samples are classified as entropy-sensitive. Consequently, APG–AMS naturally reduces to full adaptive search rather than risking aggressive pruning.

Therefore APG–AMS should be viewed as an adaptive screening mechanism whose effectiveness depends on the existence of metric-stable low-entropy regions in the feature space.

5.2 Statistical testing

Across the 80 paired folds, APG-AMS was not significantly different from Euclidean k-NN or full adaptive p-search in accuracy. For APG-AMS versus Euclidean L_2, the paired t-test gave $p = 0.417432$ and the Wilcoxon signed-rank test gave $p = 0.288623$. For APG-AMS versus full p-search, the paired t-test gave $p = 0.864007$ and the Wilcoxon test gave $p = 0.640526$.

5.3 Threshold ablation

Table 2 shows the threshold-ablation study on 70/30 splits across the eight datasets.

Table 2: Threshold ablation across eight datasets.

τ	Mean Acc.	SD Acc.	Mean Red.	SD Red.
0.70	92.475	7.450	11.434	14.114
0.75	92.540	7.481	17.769	16.823
0.80	92.364	7.383	27.196	24.818
0.85	92.369	7.269	35.945	24.648
0.90	92.473	7.317	45.086	27.524
0.92	92.473	7.317	50.505	27.794
0.95	92.598	7.113	62.711	18.183
0.98	92.307	7.876	76.200	5.227

Higher thresholds produced larger search reduction. Accuracy remained comparatively stable in this benchmark, indicating that APG entropy can be used as a control parameter for computation-accuracy tradeoffs.

5.4 Runtime scaling

Table 3 reports runtime scaling on synthetic datasets. APG-AMS gave strong runtime gains when many samples were entropy-stable, especially in lower-dimensional cases. The benefit decreased in high-dimensional settings where most samples were classified as entropy-sensitive.

Proposition 1 (Expected Search Complexity). Let

$$\alpha = \frac{\text{number of entropy-sensitive queries}}{\text{total number of queries}}$$

denote the fraction of entropy-sensitive samples.

For a candidate metric set of size five, APG–AMS performs

$$C_{\text{APG}} = N[(1-\alpha) + 5\alpha] = N(1+4\alpha)$$

distance-evaluation units.

Full adaptive search performs

$$C_{\text{Full}} = 5N.$$

Therefore

$$\frac{C_{\text{APG}}}{C_{\text{Full}}} = \frac{1+4\alpha}{5}.$$

The corresponding theoretical search reduction is

$$R = 1 - \frac{1+4\alpha}{5}.$$

Thus APG–AMS achieves maximal computational savings when α is small and naturally converges to full adaptive search as $\alpha \to 1$.

Table 3: Runtime scaling on synthetic datasets.

Samples	Features	Full Time	APG Time	Search Red.	Runtime Red.
200	10	0.008	0.001	65.333	81.512
200	30	0.022	0.014	16.000	34.392
200	60	0.036	0.034	1.333	3.147
500	10	0.040	0.024	32.533	40.272
500	30	0.113	0.106	3.733	6.229
500	60	0.235	0.213	1.067	9.369
1000	10	0.163	0.114	24.533	29.792
1000	30	0.449	0.438	2.933	2.559
1000	60	0.886	0.881	0.000	0.621

6 Discussion

The experiments support the central interpretation of APG-AMS: APG entropy is useful as a metric-stability screening variable. APG-AMS did not consistently improve accuracy over Euclidean k-NN or full adaptive p-search. Instead, it reduced the number of metric evaluations while preserving comparable accuracy.

The statistical results are important. Since APG-AMS accuracy was not significantly different from either Euclidean k-NN or full p-search, the main advantage lies in computational savings rather than predictive superiority. This is consistent with the theoretical role of entropy in APG: entropy predicts sensitivity to exponent deformation; it does not by itself learn a globally optimal classifier.

The results also reveal a limitation. In high-dimensional synthetic data, normalized entropy was frequently high, causing APG-AMS to perform nearly full search. Thus, APG-AMS is most effective when the dataset contains a substantial fraction of low-entropy, metric-stable query samples. This behavior is expected from the theory: high-entropy vectors should be treated as more sensitive to norm changes.

Compared with NCA and LMNN-type metric learning, APG-AMS addresses a different problem (Goldberger et al., 2005; Weinberger and Saul, 2009). NCA and LMNN learn transformations. APG-AMS decides when local metric search can be skipped. Therefore, APG-AMS should be viewed as a computational screening layer that may be combined with metric learning in future work.

7 Limitations

This study has several limitations.

First, the study does not include direct experimental comparisons with supervised metric-learning methods such as LMNN or NCA. Consequently, comparisons with such methods remain conceptual rather than empirical.

Second, the runtime-scaling study was constrained by available execution time and therefore used moderate synthetic problem sizes.

Third, APG–AMS was evaluated using $k = 7$ and the candidate metric set $L_1, L_{1.5}, L_2, L_3, L_4$. The effect of alternative neighborhood sizes and metric sets remains an open empirical question.

Fourth, the APG sensitivity theorem is a local result near the Euclidean reference exponent $p = 2$. APG–AMS should therefore be interpreted as an entropy-guided screening framework rather than a global replacement for exact distance optimization or supervised metric-learning methods.

8 Conclusion

This paper introduced APG-AMS, an entropy-guided adaptive metric-selection framework derived from Arithmetic Power Geometry. A d-dimensional local theorem showed that Shannon entropy governs the first-order sensitivity of normalized power-sum distortion near the Euclidean exponent. Experiments across eight datasets demonstrated that APG-AMS can reduce metric-search operations while preserving accuracy comparable to Euclidean k-NN and full adaptive p-search. The strongest supported claim is that APG entropy appears to identify metric-stable regions of feature space in the evaluated datasets and can serve as a practical screening mechanism for adaptive metric selection.

Data and Reproducibility Statement

The numerical results reported in this manuscript were produced using standard Python machine-learning libraries. The generated result files include 10-fold cross-validation summaries, fold-level outputs, threshold ablation results, and runtime-scaling outputs.

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